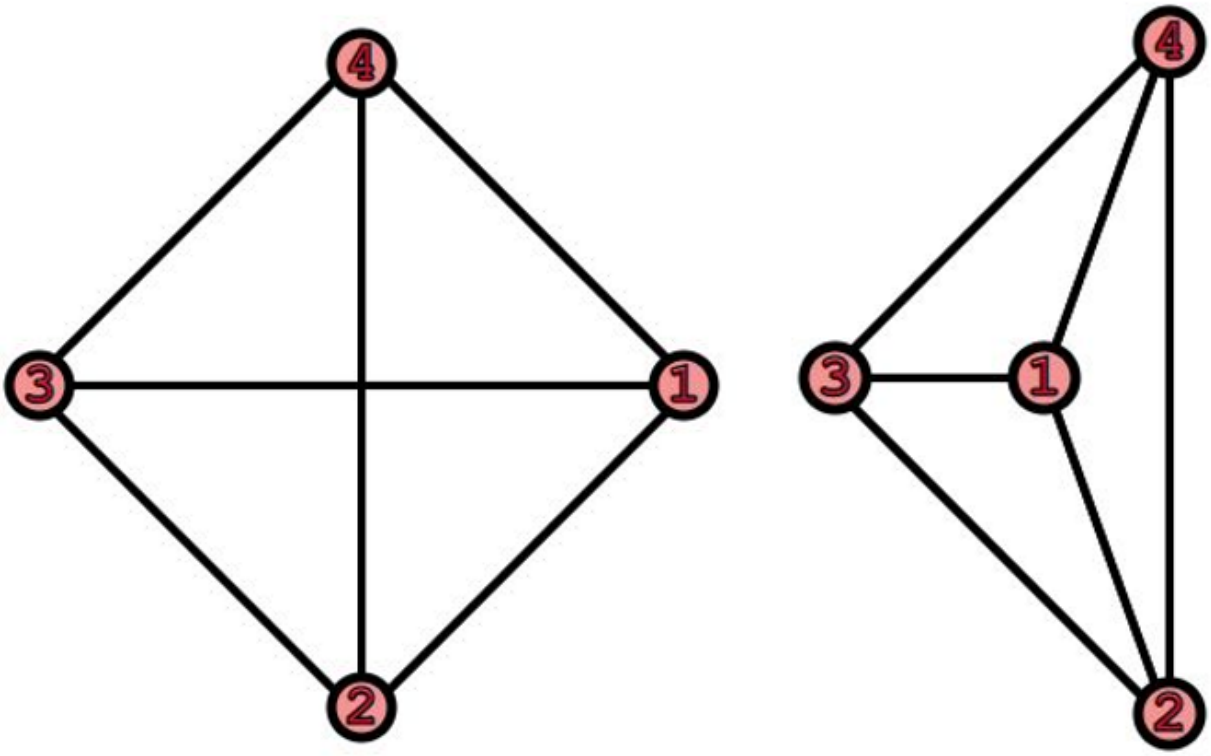


Figure 15.8 Two embeddings for K_4: although it would seem a good candidate for non-planarity, it's enough to move one vertex around to find a planar embedding, as shown on the right.

At first sight, if we draw this graph naively, it seems it has a pair of crossing edges; it's easy, however, to move one of the vertices past the crossing point to obtain a planar embedding.

The other possibility to rule out is that the is a non-planar graph with fewer edges than K_5; however, if we look at figure 15.7, it's immediately obvious that if we remove either edge 1->5 or edge 2->3, then we also get rid of the one intersection in the drawing; since a complete graph is symmetrical and relabeling-invariant[8], we can obtain an